Supplemental Materials

FDS Reconstruction of a Real Apartment Fire: Bounding Validation with Post-Fire Damage Indicators

Journal of Performance of Constructed Facilities

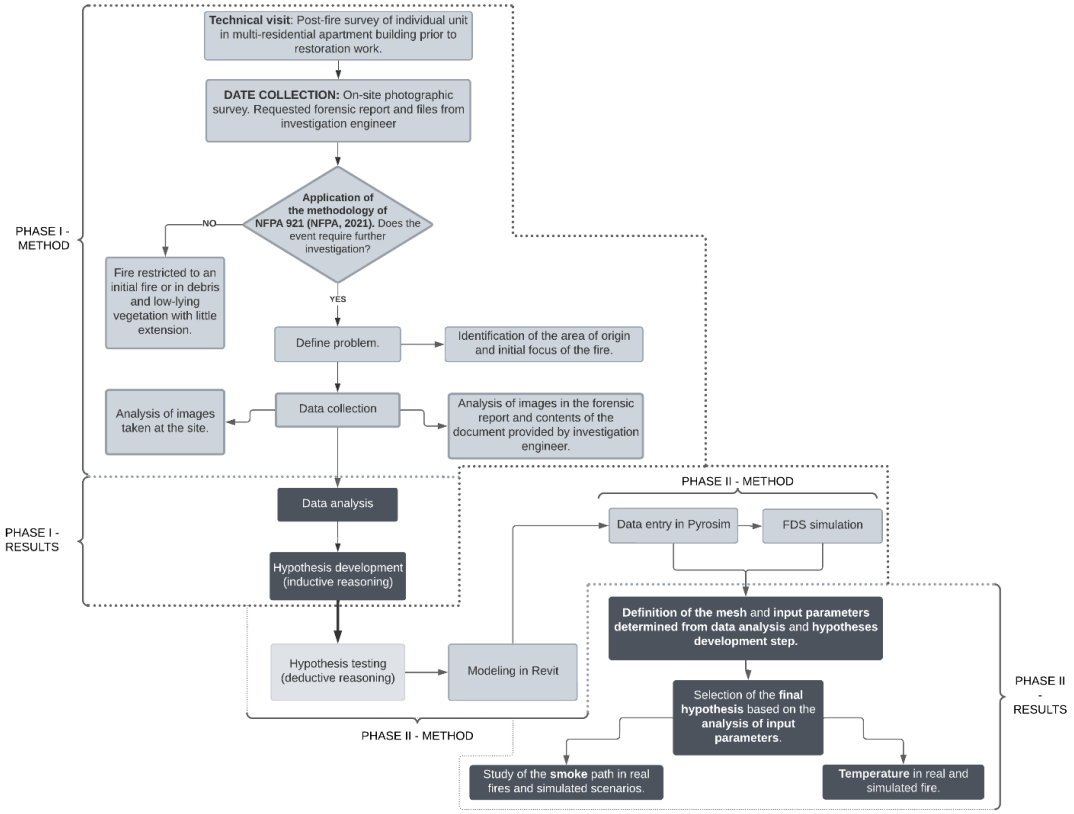


Fig. S1. Flowchart of the study: Phase I (characterization of the event and formulation of hypotheses) and Phase II (numerical modeling and hypothesis testing).

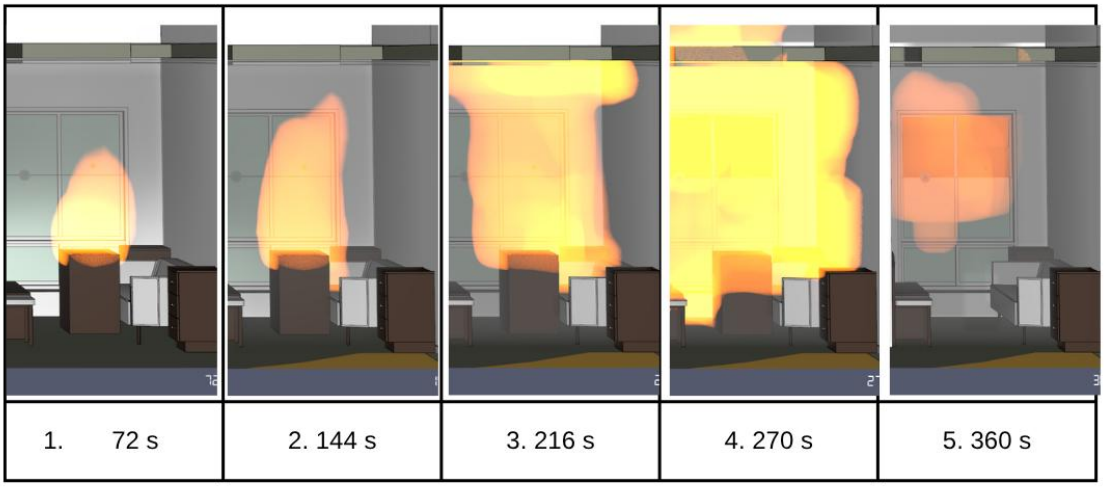


Fig. S2. Fire development in the living room obtained from FDS with the polyurethane reaction at 72, 144, 216, 270 and 360 s (smoke opacity reduced by 90% for visualization).

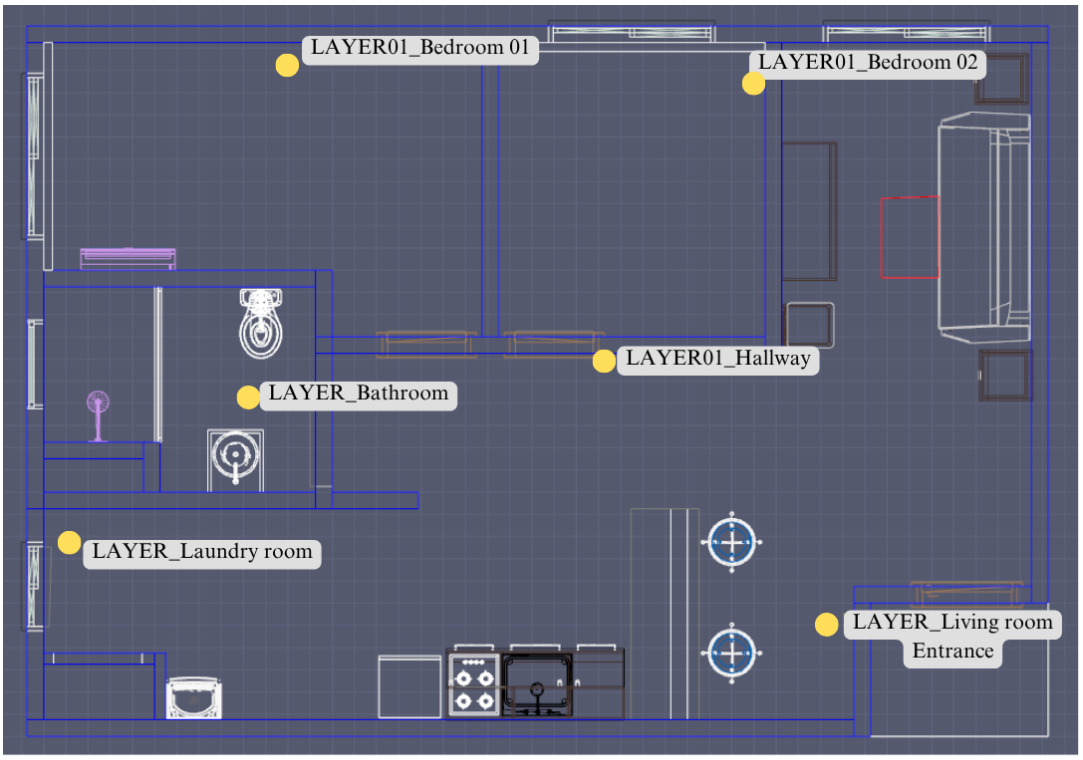


Fig. S3. Position of the layer-height devices in each room of the apartment.

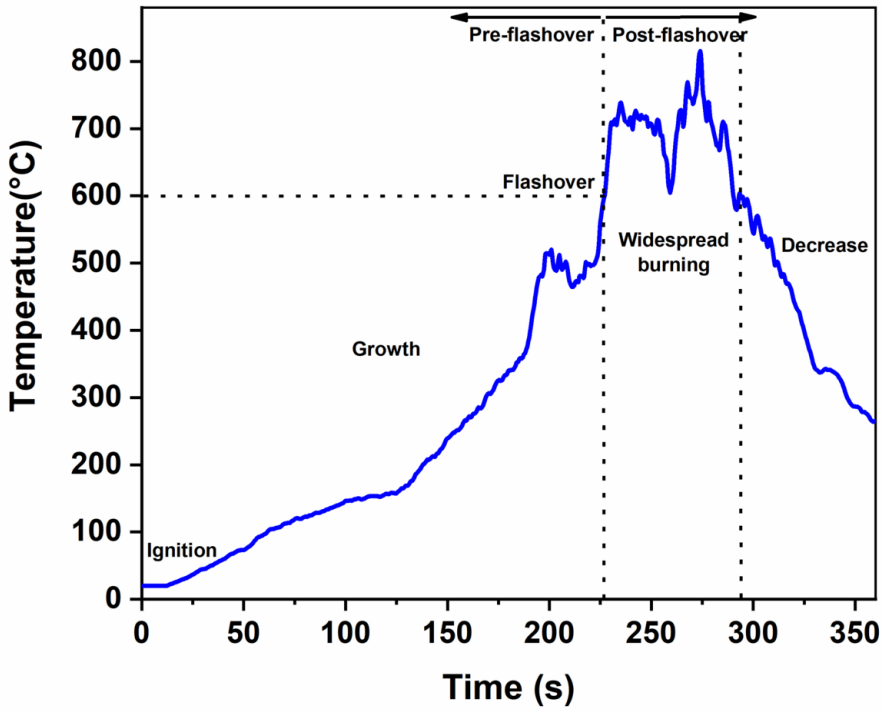


Fig. S4. Temperature at the living-room entrance (THCP, 2.10 m above the floor; 0.20-m grid), showing the growth, flashover and early decay stages.

Table S1. Surface characteristics, materials and thermal properties of the simulated elements

Surface	Material	k (W/(m·K))	c_p (kJ/(kg·K))	ρ (kg/m³)	ΔH_c (MJ/kg)	T_{ig} (°C)
White-painted structural concrete	Concrete	1.8	1.04	2,280	—	—
White paint	Fire brick	—	1.04	750	—	—
Ceiling	Gypsum plaster	0.48	0.84	1,440	—	—
PVC	PVC	—	—	1,380	—	—
Polished steel	Steel	45.8	0.46	7,850	—	—
Dark brushed aluminum	Aluminum	230	0.88	2,700	—	—
Single-pane glass	Glass	0.84	1.00	2,500	—	—
Blue plastic	Plastic	0.40	1.00	1,300	41.6	320
Dark wood	Plywood	0.15	2.30	500	8.5	220
Wood frames	Wood	0.12	1.34	400	17.5	210
Foam mat	Textile/foam	0.40	1.00	900	18.9	400

Note: Properties from Cunha and Pinto (2016), Tabaczenski et al. (2018) and the PyroSim material library (Thunderhead Engineering 2023).

Table S2. Combustion reactions tested

Reaction	Fuel formula	ΔH_c (kJ/kg)	y_{CO}	y_{soot}	y_{HCN}
SFPE POLYURETHANE_GM37	$C\ H_{1.7}\ O_{0.74}\ N_{0.002}$	1.79×10^4	0.024	0.113	0.0
NZBC C/VM2 design fuel	$C\ H_{2.202}\ O_{0.622}$	2.00×10^4	0.04	0.07	0.0

Note: Reactions from the FDS library (Babrauskas 2016; McGrattan et al. 2023a) and MBIE (2014). References are listed in the main paper.